Olympiad Test Paper — Social Science (Class 7)

Chapter 2: Understanding the Weather

Subject	Social Science (NCERT Class 7)
Chapter	2 — Understanding the Weather
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Read every stem fully — many ask you to combine several facts, read off given data, or work out a cause and its effect. Watch the classic traps: confusing weather with climate, thinking pressure rises as you climb, swapping the jobs of the barometer / thermometer / hygrometer, and assuming high humidity must mean rain. Do not write on this paper — mark answers on your answer sheet.

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1. Weather is best described as:

(A) The average pattern of the atmosphere over about thirty years
(B) The state of the Earth's atmosphere at a particular time and place
(C) The study of the instruments that measure the air
(D) The blanket of gases that surrounds the Earth

2. A geographer says, "Mumbai is hot and humid all year round, while Leh is cold and dry." This statement describes:

(A) The weather of one afternoon in each place
(B) The atmospheric pressure of each place
(C) The climate, because it is the long-term average pattern
(D) A weather forecast for the coming week

3. Almost all weather — clouds, rain, wind and storms — takes place in the:

(A) Highest, thinnest layer far above the Earth(B) Empty space beyond the atmosphere
(C) Ocean, where most of the water is stored
(D) Layer just touching the ground, called the troposphere

4. The troposphere is thicker over the warm tropics and thinner over the cold poles. The best reason is that:

(A) The poles have more water vapour than the tropics

(B) Warm air spreads out while cold air shrinks

(C) The tropics are closer to the Sun than the poles are

(D) Cold air is heavier and so pushes the layer higher

5. Which list contains ONLY elements of weather?

(A) Temperature, soil colour, rivers, wind(B) Pressure, mountains, humidity, rainfall

(C) Temperature, precipitation, pressure, wind(D) Wind, oceans, humidity, forests

6. Krishnan in Chennai messages Amir in Kashmir that the day has turned "chilly." Why is this word, by itself, not enough for Amir to know how cold it really is?

(A) "Chilly" is a personal feeling, not a fixed, shared measurement

(B) Kashmir has no thermometers to compare with

(C) The two cities are in different layers of the atmosphere

(D) "Chilly" can only describe humidity, never temperature

7. A student lists five things needed to fully describe the weather but includes one wrong item. Which item should be REMOVED?

(A) Atmospheric pressure (B) Humidity

(C) Wind (D) Soil fertility

8. We give each element of weather its own scale or unit (such as °C for temperature and mb for pressure) mainly so that:

(A) The weather changes more slowly

(B) Different people in different places can compare exact, shared values

(C) Only scientists are allowed to read the weather

(D) The instruments become cheaper to build

9. Temperature is measured with a thermometer in which many designs use a coloured liquid in a thin tube. The liquid rises when it is warmer because:

(A) The liquid expands when heated and so climbs the tube

(B) Warm liquid weighs less and floats up(C) The tube becomes wider in the heat

(D) Heat pushes air down into the tube

10. A cool day reads 15°C, which is the same as 59°F. This tells us that:

(A) 15°C is colder than 59°F

(B) Fahrenheit measures pressure, not temperature

(C) The thermometer is broken because the numbers differ

(D) Celsius and Fahrenheit are two scales used to report the same temperature

11. A city's maximum temperature one day was 32°C and its minimum was 18°C. The temperature RANGE for that day is:

(A) 50°C (B) 25°C

(C) 14°C (D) 28°C

12. Using the same day (maximum 32°C, minimum 18°C), the MEAN (average) daily temperature is:

(A) 25°C (B) 14°C
(C) 50°C (D) 32°C

13. On a summer day a town recorded a maximum of 38°C and a minimum of 26°C. Its mean daily temperature is:

(A) 12°C (B) 32°C
(C) 64°C (D) 26°C

14. Town X had a range of 6°C while town Y had a range of 15°C on the same day. This tells us that, on that day:

(A) Town X was hotter than town Y (B) Town Y had a higher average temperature
(C) Town X recorded more rainfall
(D) Town Y's temperature swung more between its high and low than town X's

15. Precipitation means:

(A) Any form of water — rain, snow, sleet or hail — that falls from the sky
(B) Only the rain that falls in the monsoon(C) The water vapour invisible in the air
(D) The weight of the air pressing down on us

16. Match the form of precipitation to its description: small, hard balls of ice that fall from the sky are called:

(A) Sleet (B) Snow
(C) Hail (D) Dew

17. Rainfall is measured with a:

(A) Barometer (B) Rain gauge
(C) Hygrometer (D) Anemometer

18. A rain gauge reading says the area got "5 mm of rainfall." This means that:

(A) Exactly 5 millilitres of rain were collected(B) The rain fell for 5 minutes
(C) The temperature dropped by 5 degrees
(D) The rain would lie 5 millimetres deep on flat ground

19. If snow, rather than rain, falls into a rain gauge, the correct step before reading the scale is to:

(A) Let the snow melt first, then measure the depth of water
(B) Pour out the snow and report zero (C) Weigh the snow on a balance instead
(D) Double the depth of the snow

20. Atmospheric pressure is best described as:

(A) The amount of water vapour in the air(B) The speed at which the wind is blowing
(C) The weight of all the air above and around us pressing down

(D) The temperature of the upper atmosphere

21. As a climber goes higher up a tall mountain, the atmospheric pressure:

(A) Increases, because the climber is closer to the sky

(B) Stays exactly the same all the way up(C) Suddenly drops to zero at the summit

(D) Decreases, because less air is left above the climber

22. A child argues, "On a mountain you are nearer the top of the sky, so the pressure must be higher up there." The mistake is that pressure depends on:

(A) How close you are to the Sun

(B) The weight of the air ABOVE you, which is less up high

(C) The temperature of the rock you stand on(D) How much it has rained recently

23. Atmospheric pressure is measured with a barometer in units called:

(A) Degrees Celsius (B) Millimetres

(C) Millibars (D) Percent

24. Normal pressure at the sea coast is about 1013 mb. A "depression," which can grow into a storm, is a region where pressure:

(A) Drops dramatically, below about 1000 mb(B) Rises far above 1013 mb

(C) Stays fixed at exactly 1013 mb (D) Cannot be measured at all

25. Wind is best defined as:

(A) Water vapour rising from the sea

(B) Air moving from areas of high pressure to areas of low pressure

(C) The weight of the air pressing down

(D) Air moving from low pressure to high pressure

26. Complete the rule: "Wind blows from areas of _____ pressure toward areas of _____ pressure."

(A) low; high (B) warm; cold

(C) wet; dry (D) high; low

27. Two regions lie side by side: region P is HIGH pressure and region Q is LOW pressure. The wind between them will blow:

(A) From P toward Q (B) From Q toward P

(C) In a circle without going anywhere (D) Only straight upward

28. If the difference in pressure between two nearby regions is made much larger, the wind between them will most likely:

(A) Stop blowing (B) Reverse its direction

(C) Blow faster (D) Turn into rain

29. The two things we measure to describe a wind are its:

(A) Colour and weight (B) Speed and direction
(C) Temperature and humidity (D) Depth and pressure

30. A wind vane has a pointer at one end and a tail at the other. When wind blows, the tail is pushed away so that the pointer:

(A) Shows how fast the wind is blowing (B) Measures the air's humidity
(C) Counts the raindrops falling
(D) Shows the direction from which the wind comes

31. Which instrument measures wind SPEED by counting how fast its rotating cups spin?

(A) Wind vane (B) Barometer
(C) Anemometer (D) Hygrometer

32. At an airport, a "wind sock" is used mainly to:

(A) Help pilots judge the wind's direction and rough strength at a glance
(B) Catch and measure rainfall (C) Measure the air pressure on the runway
(D) Record the temperature of the tarmac

33. A weather station needs to record BOTH which way the wind comes from and how fast it is. The correct pair of instruments is:

(A) A barometer and a thermometer (B) A wind vane and an anemometer
(C) A rain gauge and a hygrometer (D) Two thermometers

34. Humidity refers to:

(A) The weight of the air above us (B) The speed of the wind
(C) The depth of rainfall collected
(D) The amount of water vapour present in the air

35. Relative humidity is usually given as a percentage. A reading of 100% would mean:

(A) The air is completely full of water vapour(B) The air is completely dry
(C) It is raining heavily right now (D) The temperature has reached 100°C

36. Dry weather usually shows a relative humidity of about 20%–40%, while humid weather is about 60%–80%. So a reading of 75% best describes air that is:

(A) Very dry and crisp (B) Below freezing
(C) Damp and sticky (D) Free of all water vapour

37. Wet clothes hung out to dry will dry MOST SLOWLY when the relative humidity is:

(A) Very low, such as 25% (B) Very high, such as 90%
(C) Exactly 0%
(D) The same speed no matter what the humidity is

38. Wet clothes dry faster in Jaipur (low humidity) than in Kochi (high humidity) because, where humidity is high, the air:

(A) Is too cold to let water evaporate

(B) Has too much pressure pressing the water in

(C) Is moving too fast over the clothes

(D) Is already nearly full of vapour, so it accepts very little more from the clothes

39. The instrument used to measure humidity is the:

(A) Hygrometer (B) Barometer

(C) Anemometer (D) Rain gauge

40. A student claims, "A barometer and a thermometer measure pretty much the same thing." The correct reply is that:

(A) They are the same; both measure temperature

(B) Both measure humidity

(C) A barometer measures pressure while a thermometer measures temperature — two different elements

(D) A barometer measures rainfall and a thermometer measures wind

41. A muggy day shows very high humidity, yet it does not rain. The best explanation is that:

(A) High humidity always brings rain, so the day is unusual

(B) Rain falls only when the air cools enough for the vapour to condense; high humidity alone is not enough

(C) The hygrometer must be broken

(D) Humidity and rain have nothing to do with each other

42. Before modern instruments existed, people forecast the weather mainly by:

(A) Reading printed forecasts in newspapers(B) Using satellites and computers

(C) Measuring pressure with barometers

(D) Carefully watching the behaviour of animals and plants

43. Which of these is a traditional "Nature's clue" once used to sense that rain or a storm was coming?

(A) Ants carrying their eggs to higher ground(B) A rising barometer reading

(C) A digital thermometer beeping (D) A weather satellite passing overhead

44. Animals and plants can react to coming rain before people notice it because they:

(A) Are smarter than meteorologists (B) Can read the barometer themselves

(C) Are exposed to the air all the time, so changes in pressure and humidity affect their bodies and behaviour

(D) Always know the date the monsoon will arrive

45. A real weakness of relying only on "Nature's clues" to forecast weather is that the clues are:

(A) Always perfectly accurate

(B) Tied to one place and give only vague, short warnings like "rain is coming"

(C) Measured precisely in numbers

(D) Gathered from across the whole country at once

46. Meteorology is best defined as the:

(A) Study of rocks and soil (B) Study of the oceans and tides

(C) Art of guessing the weather from the sky

(D) Systematic study of weather and how it changes

47. How does meteorology differ from simply reading Nature's clues?

(A) Meteorology measures each element precisely and gathers data over time, instead of relying on a single vague sign

(B) Meteorology uses only one animal as its guide

(C) Meteorology ignores all instruments

(D) Meteorology can only describe today, never predict ahead

48. A child says, "Meteorologists just look at the sky and guess." The accurate correction is that they:

(A) Do indeed only guess (B) Only read the behaviour of frogs and ants

(C) Use instruments to measure each element, collect data from many places over time, and apply scientific methods

(D) Rely entirely on newspaper reports

49. A weather station is useful chiefly because it:

(A) Removes the need for any instruments(B) Predicts the weather without any data

(C) Can only measure rainfall

(D) Brings all the different weather instruments together in one place, read at regular times

50. An Automated Weather Station (AWS) differs from an ordinary weather station mainly in that it:

(A) Cannot measure rainfall

(B) Uses sensors to measure and record data by itself, without a person present

(C) Only works during the daytime (D) Measures climate instead of weather

51. Why is an Automated Weather Station especially useful in remote or risky places?

(A) It can keep recording on its own where it is hard or dangerous for people to stay

(B) It is cheaper than a thermometer (C) It needs an operator to read it every hour

(D) It only works inside cities

52. To bring together a thermometer, rain gauge, barometer, wind vane, anemometer and hygrometer in one place, and read them at set times, you would set up a:

(A) Climate map
(B) Wind sock
(C) Weather station
(D) Depression

53. The chief reason accurate weather forecasts matter so much is that they:

(A) Make the weather change less often

(B) Give people and governments time to prepare for floods, cyclones, droughts and heat waves

(C) Replace the need for any instruments(D) Lower the air pressure during storms

54. In India, the official body that gathers weather data and issues forecasts and warnings is the:

(A) National Rain Gauge Office
(B) Department of Climate History
(C) Indian Wind Vane Society
(D) India Meteorological Department

55. A fisherman is told by a forecast not to sail today because a storm is expected. This shows that forecasts are valuable because they:

(A) Warn people in advance so they can stay safe

(B) Stop the storm from forming
(C) Measure how deep the sea is

(D) Increase the day's humidity

56. Over a few hours a weather station records the barometer FALLING, the hygrometer RISING, the wind SPEEDING UP and the sky CLOUDING over. The best forecast is:

(A) Clear, calm, dry weather
(B) Wet, stormy weather is on the way

(C) A drop in temperature with no clouds(D) Nothing will change at all

57. In the same situation, the FALLING barometer is an early sign of trouble because falling pressure often marks:

(A) Perfectly settled, clear weather
(B) A rise in the mountain's height

(C) The arrival of a depression, the seed of a storm

(D) The end of all wind

58. A weather report says a hill station read 650 mb while a nearby coastal town read 1013 mb at the same time. The most sensible explanation is that:

(A) The hill station's barometer is broken(B) The coast is colder than the hill station

(C) The hill station had heavier rain

(D) Pressure is lower at the hill station because it is higher up, with less air above it

59. A traveller climbing to a high pass feels breathless and dizzy. Linking this to the chapter, the best reason is that high up the air is:

(A) Heavier, with more oxygen than at the coast

(B) Thinner, with lower pressure and less oxygen

(C) Completely free of any gases
(D) Full of rain that blocks breathing

60. Putting it all together, which pairing of weather element and its instrument is INCORRECT?

(A) Temperature → thermometer
(B) Rainfall → rain gauge
(C) Humidity → barometer
(D) Wind speed → anemometer

End of question paper. Maximum marks: 240. Marking: +4 correct, −1 wrong, 0 unattempted.

Solutions — Social Science (Class 7), Chapter 2: Understanding the Weather

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	D	31	C	41	B	51	A
2	C	12	A	22	B	32	A	42	D	52	C
3	D	13	B	23	C	33	B	43	A	53	B
4	B	14	D	24	A	34	D	44	C	54	D
5	C	15	A	25	B	35	A	45	B	55	A
6	A	16	C	26	D	36	C	46	D	56	B
7	D	17	B	27	A	37	B	47	A	57	C
8	B	18	D	28	C	38	D	48	C	58	D
9	A	19	A	29	B	39	A	49	D	59	B
10	D	20	C	30	D	40	C	50	B	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Weather is the state of the Earth's atmosphere at a particular time and place — hot or cold, dry or rainy, calm or windy, right now and right here. (D) defines the atmosphere itself, not the weather; the "thirty-year average" option describes climate, not weather.

2. (C) A statement about how a place is "all year round" is about the long-term average pattern — that is climate, not weather. Weather would be one moment's conditions (an afternoon), so the "one afternoon" option is the weather trap, and a forecast is only about the near future.

3. (D) Almost all weather happens in the troposphere, the lowest layer of the atmosphere touching the ground, because the air, warmth and water vapour needed to make clouds and rain are concentrated there. The highest, thinnest layers are too dry; space and the ocean are not where atmospheric weather forms.

4. (B) Warm air spreads out (expands) so the troposphere is thicker over the warm tropics, while cold air shrinks so it is thinner over the cold poles. Being "closer to the Sun" is not the textbook reason, and cold air shrinking — not pushing the layer higher — is what actually happens at the poles.

5. (C) The five elements of weather are temperature, precipitation, atmospheric pressure, wind and humidity. Only option (C) lists four of these and nothing else. Each of the other lists smuggles in non-weather items such as soil colour, rivers, mountains, oceans or forests.

6. (A) "Chilly" is a personal feeling — what feels chilly to someone used to warm Chennai may feel mild to someone in cold Kashmir. Without a fixed measurement (a thermometer reading in °C), Amir cannot know the true temperature. The cities are in the same layer (the troposphere), and the word is about temperature, not humidity.

7. (D) Temperature, precipitation, atmospheric pressure, wind and humidity are the five elements; soil fertility is not a weather element at all, so it is the wrong item to remove from a true list. Pressure, humidity and wind all genuinely belong.

8. (B) A vague word like "hot" means different things to different people; a shared scale (°C, mb) turns a feeling into an exact number that anyone, anywhere, can compare. Standard units do not slow the weather down or make instruments cheaper, and they let everyone — not only scientists — read the values.

9. (A) Most liquids expand when heated, so the coloured liquid takes up more room and climbs the thin tube; it sinks when cooled. The liquid does not "float up" by weighing less, the glass tube does not widen, and no air is pushed down — those are made-up mechanisms.

10. (D) 15°C and 59°F are the SAME temperature written on two different scales (Celsius and Fahrenheit), so neither is colder and the thermometer is not broken. Fahrenheit measures temperature, not pressure.

11. (C) Range = maximum − minimum = 32 − 18 = 14°C. The 50 trap adds the two readings, 25 is the mean (their average), and 28 has no basis.

12. (A) Mean daily temperature = (maximum + minimum) ÷ 2 = (32 + 18) ÷ 2 = 50 ÷ 2 = 25°C. The 14 trap is the range (the difference), 50 is the un-halved sum, and 32 is just the maximum.

13. (B) Mean = (38 + 26) ÷ 2 = 64 ÷ 2 = 32°C. The 12 trap is the range (38 − 26), 64 is the sum before halving, and 26 is just the minimum.

14. (D) A larger range means a bigger swing between the day's high and low. Town Y's range (15°C) is bigger than town X's (6°C), so Y's temperature swung more. Range says nothing about which town was hotter, about average temperature, or about rainfall.

15. (A) Precipitation is ANY form of water that falls from the sky — rain, snow, sleet or hail — not only monsoon rain. Invisible water vapour stays in the air (so it has not fallen), and the weight of air is pressure, not precipitation.

16. (C) Hail is small, hard balls of ice that fall from the sky. Sleet is frozen or partly frozen rain, snow is ice crystals, and dew forms on surfaces from condensation rather than falling as ice balls.

17. (B) Rainfall is measured with a rain gauge: rain enters a funnel, collects in a measuring tube, and a scale reads the depth. A barometer measures pressure, a hygrometer humidity, and an anemometer wind speed.

18. (D) A rain-gauge reading is a DEPTH: "5 mm of rainfall" means the rain would lie 5 millimetres deep over flat ground. It is not 5 millilitres collected, not 5 minutes of rain, and has nothing to do with temperature.

19. (A) Snow must be melted to liquid water first, because the gauge's scale measures the depth of water, not loose snow. Pouring it out (reporting zero), weighing it, or doubling the snow depth would all give a wrong rainfall figure.

20. (C) Atmospheric pressure is the weight of all the air above and around us pressing down. Water vapour is humidity, the speed of moving air is wind, and upper-atmosphere temperature is a different element entirely.

21. (D) Pressure is the weight of the air above you; the higher you climb, the less air is left above, so pressure decreases. It does not increase with height, it does not stay constant, and it falls gradually rather than dropping to zero at the summit.

22. (B) The child's error is forgetting that pressure depends on the weight of the air ABOVE you — and high up there is less air above, so pressure is lower, not higher. It does not depend on nearness to the Sun, the rock's temperature, or recent rain.

23. (C) A barometer measures atmospheric pressure in millibars (mb); normal coastal pressure is about 1013 mb. Degrees Celsius are for temperature, millimetres for rainfall depth, and percent for relative humidity.

24. (A) A depression is a low-pressure region where pressure drops dramatically — below about 1000 mb — and can develop into a storm or cyclone. Rising far above normal would be high pressure, staying fixed describes no change, and depressions are certainly measurable on a barometer.

25. (B) Wind is moving air, and air flows from areas of HIGH pressure to areas of LOW pressure (like air rushing out of a balloon). Rising vapour and the weight of air are different things, and "low to high" is exactly backwards.

26. (D) Air slides "downhill" in pressure: from high pressure toward low pressure. So the blanks are high; low. "Low; high" reverses it, and warm/cold or wet/dry are not the rule that defines wind direction.

27. (A) Wind blows from the high-pressure region to the low-pressure region, so it goes from P (high) toward Q (low). "Q toward P" reverses the rule; wind between two such regions flows across, not in a closed circle or straight up.

28. (C) The bigger the pressure difference, the faster the wind blows — so a much larger difference makes the wind speed up. A bigger difference would not stop the wind, reverse it, or turn it into rain.

29. (B) A wind is described by its speed (how fast) and its direction (where it comes from). Colour, weight, temperature, humidity, depth and pressure are not the two quantities used to describe a wind.

30. (D) When wind pushes the tail, the pointer swings to show the direction FROM which the wind is coming — that is the job of a wind vane. Wind speed is the anemometer's job, humidity the hygrometer's, and counting raindrops belongs to no instrument here.

31. (C) An anemometer has rotating cups that spin faster in stronger wind, and a meter counts the spins to give the wind speed. A wind vane shows direction, a barometer pressure, and a hygrometer humidity.

32. (A) A wind sock at an airport lets a pilot judge the wind's direction and rough strength at a glance during take-off and landing. It does not measure rainfall, pressure or temperature.

33. (B) Direction needs a wind vane and speed needs an anemometer, so the pair is wind vane + anemometer. A barometer/thermometer pair measures pressure and temperature, a rain gauge/hygrometer pair rainfall and humidity, and two thermometers measure neither aspect of wind.

34. (D) Humidity is the amount of water vapour present in the air, usually given as relative humidity (a percentage). The weight of air is pressure, the speed of air is wind, and collected rainfall depth is precipitation.

35. (A) Relative humidity of 100% means the air is completely full of water vapour and can hold no more. 0% would be completely dry; 100% humidity does not by itself mean rain is falling, and it has nothing to do with a 100°C temperature.

36. (C) Humid weather runs about 60%–80%, so 75% sits in the humid band: damp and sticky. Dry/crisp air is 20%–40%; "below freezing" is about temperature, not humidity; and 75% is far from free of vapour.

37. (B) When relative humidity is very high (e.g. 90%), the air is already nearly full of vapour, so it accepts very little more and wet clothes dry most slowly. At very low humidity they dry fast; 0% never really happens; and drying speed clearly does depend on humidity.

38. (D) Where humidity is high, the air is already nearly full of water vapour, so it can take in very little more from the wet clothes — they dry slowly. It is not about the air being too cold, pressure pressing water in, or the wind speed.

39. (A) Humidity is measured with a hygrometer. A barometer measures pressure, an anemometer wind speed, and a rain gauge rainfall.

40. (C) A barometer and a thermometer measure two DIFFERENT elements: the barometer measures atmospheric pressure (the weight of the air) and the thermometer measures temperature (how hot or cold the air is). They are not the same, neither measures humidity, and neither measures rainfall or wind.

41. (B) High humidity only means the air holds a lot of vapour; rain falls only when that air cools enough to reach its dew point so the vapour condenses. So a muggy day can stay rainless — the hygrometer is fine, and humidity and rain are clearly related, just not the same thing.

42. (D) Before modern instruments, people forecast weather by carefully watching the behaviour of animals and plants (ants, frogs, birds, pine cones). Newspapers, satellites/computers and barometers are all modern tools, not the old way.

43. (A) Ants carrying their eggs to higher ground is a classic traditional "Nature's clue" warning of coming rain. A barometer reading, a digital thermometer and a weather satellite are all modern instruments, not natural signs.

44. (C) Animals and plants are exposed to the air all the time, so the changes in pressure and humidity that come before rain affect their bodies and behaviour, making them react early. They are not "smarter," they cannot read instruments, and they do not know exact dates.

45. (B) Nature's clues are tied to one place and give only vague, short warnings such as "rain is coming" — that is their real weakness. They are not always accurate, not measured precisely in numbers, and not gathered nationwide at once (that is what meteorology adds).

46. (D) Meteorology is the systematic study of weather and how it changes, and it is the basis of forecasting. It is not the study of rocks/soil or oceans/tides, and it is the opposite of mere guessing.

47. (A) Meteorology measures each element precisely and gathers data from many places over time, instead of relying on a single vague natural sign. It does not use only one animal, it depends on instruments rather than ignoring them, and it does predict ahead, not just describe today.

48. (C) Meteorologists do not guess: they use instruments to measure each element, collect data from many places over time, and apply scientific methods to predict the weather. They do not rely only on frogs and ants or on newspaper reports.

49. (D) A weather station is useful because it brings all the different instruments (thermometer, rain gauge, barometer, wind vane, anemometer, hygrometer) together in one place, read at regular times. It does not remove the need for instruments, cannot predict without data, and measures more than just rainfall.

50. (B) An Automated Weather Station (AWS) uses sensors to measure and record weather data by itself, without a person present. It can measure rainfall, can work day and night, and still measures weather (not climate).

51. (A) Because an AWS records on its own, it is ideal for remote or risky places where it is hard or dangerous for people to stay. It is not chosen for being cheaper than a thermometer, it does NOT need an hourly operator (that is the opposite of automated), and it is not limited to cities.

52. (C) Gathering all those instruments in one place and reading them at set times is exactly what a weather station does. A climate map shows long-term averages, a wind sock shows wind at an airport, and a depression is a low-pressure region — none of these gather the instruments.

53. (B) Accurate forecasts matter because they give people and governments time to prepare for floods, cyclones, droughts and heat waves. Forecasts do not make the weather change less, do not replace instruments, and do not alter the air pressure during storms.

54. (D) In India the India Meteorological Department gathers weather data and issues forecasts and warnings. The other three are invented names with no such role.

55. (A) Telling a fisherman not to sail because a storm is coming shows the value of forecasts: they warn people in advance so they can stay safe. Forecasts cannot stop a storm forming, do not measure sea depth, and do not change the humidity.

56. (B) A falling barometer (pressure dropping), rising humidity, strengthening wind and clouding sky all point the same way: wet, stormy weather is on the way. Clear/calm/dry weather, a cloudless temperature drop, or "no change" all contradict the clues.

57. (C) Falling pressure often marks the arrival of a depression — a low-pressure system that is the seed of a storm — which is why a dropping barometer is an early warning. It does not signal settled clear weather, a change in altitude, or the end of all wind.

58. (D) The hill station reads lower pressure (650 mb) than the coast (1013 mb) because it is higher up, with less air above it pressing down — exactly as pressure should behave with altitude. The barometer is not broken, the reading is not explained by the coast being colder, and heavier rain does not cause that gap.

59. (B) High up the air is thinner, so both the pressure and the amount of oxygen are lower, which is why climbers feel breathless and dizzy. The air is not heavier with more oxygen, it is not free of all gases, and the breathlessness is not caused by rain.

60. (C) Humidity is measured with a hygrometer, NOT a barometer (which measures pressure), so "Humidity → barometer" is the incorrect pairing. Temperature → thermometer, rainfall → rain gauge, and wind speed → anemometer are all correct.

Solutions complete: 60 / 60.